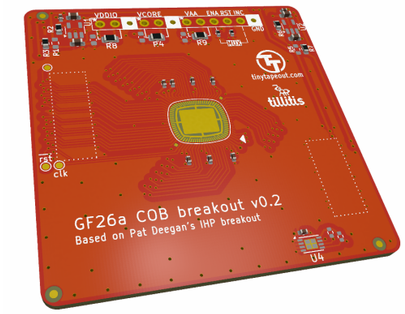
QFN and Analog Breakout For TinyTapeout GF CoB ASICs

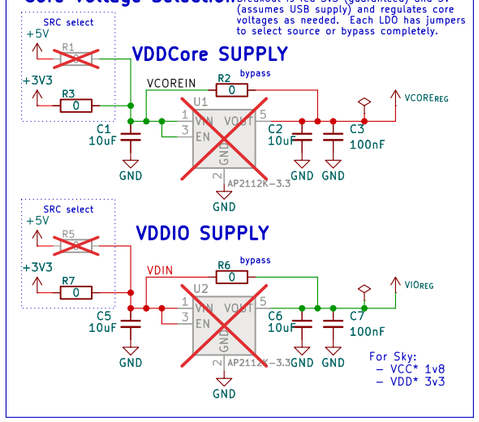
This version of the breakout allows access to the ASIC I/O, management and analog interfaces.
It is designed for wirebonding GF die ASICs to demoboard breakout, and

* it receives the DB 3v3 IO level as well as the VUSB and it takes on the responsibility of regulating its IO/Vcore voltages
* provides interfaces to supply or monitor each VDD* and VAA independently
* provides means to send off analog signals to demoboard, and its weak pull-downs, or severe that connection
* is naturally keyed, owing to new bottom-side connectors
* DF12 connectors have high speed capability (up to USB4/20Gbps), high density and, optionally, a shielded version for improved noise immunity.
* has additional I/O from bottom-side for management, eeprom ID or expansion.

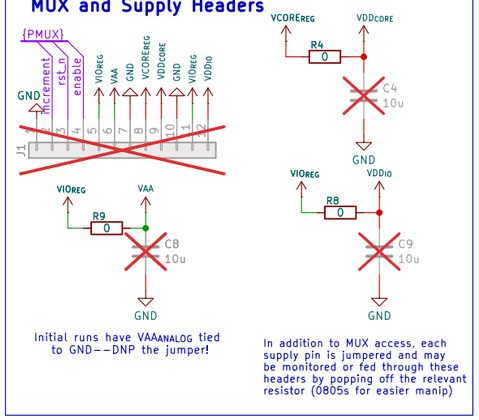


Core Voltage Selection

Breakout is fed 3v3 (guaranteed) and 5V (assumes USB supply) and regulates core voltages as needed. Each LDO has jumpers to select source or bypass completely.

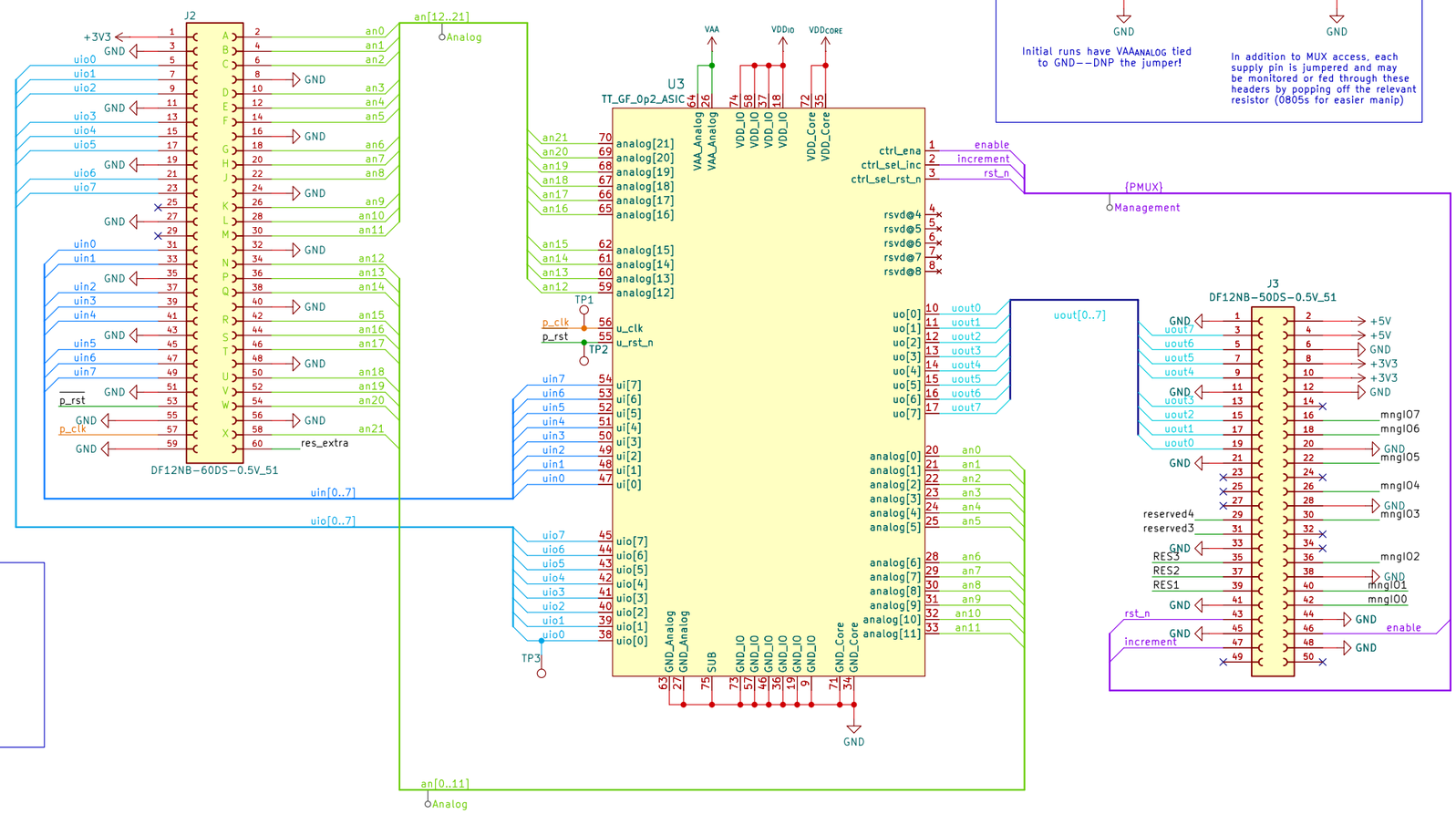


MUX and Supply Headers



Initial runs have VAA_{ANALOG} tied to GND--DNP the jumper!

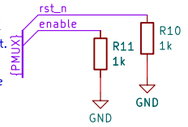
In addition to MUX access, each supply pin is jumpered and may be monitored or fed through these headers by popping off the relevant resistor (0805s for easier manip)



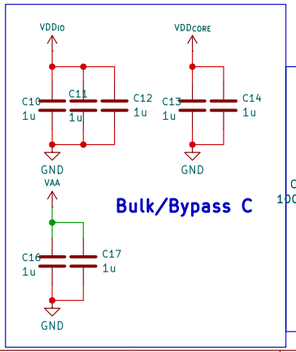
Presence Detection

pull-downs used in conjunction with weak PU on demoboard for poor man's ASIC detector circuit.

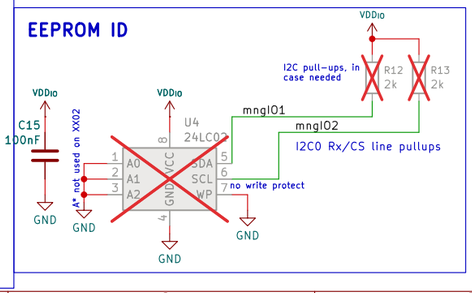
Other possibility is to use a dedicated i2c eeprom and probe for that.



Bulk/Bypass C



EEPROM ID



FID1 Fiducial
FID2 Fiducial
FID3 Fiducial

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Psychogenic Technologies INC

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File: breakout-gf-cob.kicad_sch

Title: Breakout For TinyTapeout GF Cob ASICs

Size: A3	Date: 2026-10-28	**Rev: 1.2**
KiCad E.D.A. 10.0.7		Id: 1/1